WORKED SOLUTION

STEP III 2001, Question 10

Mechanics • Relative Velocity

> **STEP III 2001, Question 10**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/01-s3-q10

The key idea

Relative velocities are differences: the velocity of A relative to C is $\mathbf{v}_1 - \mathbf{u}$. Once both conditions are written as dot products, everything is measured from the midpoint $\frac{1}{2}(\mathbf{v}_1 + \mathbf{v}_2)$, and the two conditions become a difference of two squares. Read geometrically, the equations describe a circle and a line, and that is where the count of two velocities comes from.

Solution

The conditions

We treat the ships as particles moving in the horizontal plane of the sea, so all the velocities are two-dimensional vectors. The velocities of A and B relative to C are

$$\mathbf{a} = \mathbf{v}_1 - \mathbf{u}, \qquad \mathbf{b} = \mathbf{v}_2 - \mathbf{u}.$$

Equal magnitudes and perpendicularity mean

$$|\mathbf{v}_1 - \mathbf{u}|^2 = |\mathbf{v}_2 - \mathbf{u}|^2 \qquad \text{and} \qquad (\mathbf{v}_1 - \mathbf{u}) \cdot (\mathbf{v}_2 - \mathbf{u}) = 0.$$

The two equations

Write

$$\mathbf{w} = \mathbf{u} - \tfrac{1}{2}(\mathbf{v}_1 + \mathbf{v}_2), \qquad \mathbf{d} = \tfrac{1}{2}(\mathbf{v}_1 - \mathbf{v}_2).$$

Then $\mathbf{v}_1 = \frac{1}{2}(\mathbf{v}_1 + \mathbf{v}_2) + \mathbf{d}$ and $\mathbf{v}_2 = \frac{1}{2}(\mathbf{v}_1 + \mathbf{v}_2) - \mathbf{d}$, so

$$\mathbf{a} = \mathbf{d} - \mathbf{w}, \qquad \mathbf{b} = -\mathbf{d} - \mathbf{w}.$$

Now expand each condition, using $\mathbf{w} \cdot \mathbf{d} = \mathbf{d} \cdot \mathbf{w}$:

$$\mathbf{a} \cdot \mathbf{b} = -(\mathbf{d} - \mathbf{w}) \cdot (\mathbf{d} + \mathbf{w}) = |\mathbf{w}|^2 - |\mathbf{d}|^2,$$
$$|\mathbf{a}|^2 - |\mathbf{b}|^2 = (|\mathbf{d}|^2 - 2\mathbf{d} \cdot \mathbf{w} + |\mathbf{w}|^2) - (|\mathbf{d}|^2 + 2\mathbf{d} \cdot \mathbf{w} + |\mathbf{w}|^2) = -4\,\mathbf{w} \cdot \mathbf{d}.$$

So perpendicularity, $\mathbf{a}\cdot\mathbf{b}=0$, says exactly $|\mathbf{w}|^2=|\mathbf{d}|^2$, which is

$$\left|\mathbf{u}-\tfrac{1}{2}(\mathbf{v}_1+\mathbf{v}_2)\right|^2=\left|\tfrac{1}{2}(\mathbf{v}_1-\mathbf{v}_2)\right|^2,$$

and equal magnitudes, $|\mathbf{a}|^2=|\mathbf{b}|^2$, says exactly $\mathbf{w}\cdot\mathbf{d}=0$, which (multiplying by 2) is

$$\Big(\mathbf{u}-\tfrac{1}{2}(\mathbf{v}_1+\mathbf{v}_2)\Big)\cdot(\mathbf{v}_1-\mathbf{v}_2)=0.$$

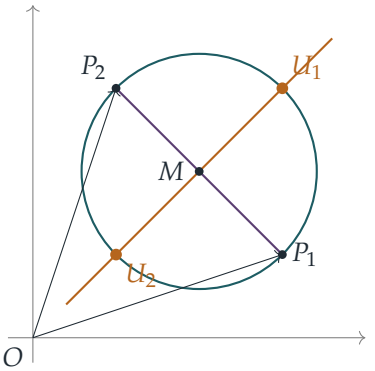
These are the two required equations. Each step above was an identity, so the argument also runs backwards: the two equations hold *if and only if* the two conditions hold.

Why there are two velocities for C

Draw velocities as position vectors from a common origin, and let P_1, P_2, U be the tips of $\mathbf{v}_1$, $\mathbf{v}_2$, $\mathbf{u}$, with M the midpoint of P_1P_2.

Ask yourself What does each equation say about where the point U can be?

The first equation says U is at distance $\frac{1}{2}|\mathbf{v}_1-\mathbf{v}_2|$ from M: it lies on the circle with diameter P_1P_2. The second says MU is perpendicular to P_1P_2 (or $U=M$): U lies on the perpendicular bisector of P_1P_2.



In the plane, a line through the centre of a circle of positive radius meets it in exactly two points, at opposite ends of a diameter. The radius is $\frac{1}{2}|\mathbf{v}_1-\mathbf{v}_2|$, which is positive precisely because $\mathbf{v}_1\neq\mathbf{v}_2$. So there are exactly two possible velocities for C:

$$\mathbf{u}=\tfrac{1}{2}(\mathbf{v}_1+\mathbf{v}_2)\pm\tfrac{1}{2}\,\mathbf{d}^{\perp},$$

where $\mathbf{d}^{\perp}$ is $\mathbf{v}_1-\mathbf{v}_2$ turned through a right angle. (If $\mathbf{v}_1=\mathbf{v}_2$ the circle shrinks to the single point M, and then both relative velocities are zero, so the conditions say nothing useful.) In the diagram, the right angle P_1UP_2 is the angle in a semicircle, and $UP_1=UP_2$ because U is on the perpendicular bisector.

The final part

Now $|\mathbf{v}_1| = |\mathbf{v}_2| = k$ and $\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$, where $k > 0$ (perpendicularity only makes sense for non-zero vectors). Then $\mathbf{v}_1$ and $\mathbf{v}_2$ are perpendicular and non-zero, so they span the plane, and we may write $\mathbf{u} = \alpha\mathbf{v}_1 + \beta\mathbf{v}_2$.

The second equation, using $\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$ and $|\mathbf{v}_1|^2 = |\mathbf{v}_2|^2$, gives

$$\mathbf{u} \cdot (\mathbf{v}_1 - \mathbf{v}_2) = \tfrac{1}{2}\left(|\mathbf{v}_1|^2 - |\mathbf{v}_2|^2\right) = 0, \qquad \text{so} \qquad \alpha k^2 - \beta k^2 = 0,$$

and hence $\beta = \alpha$. For the first, expand:

$$|\mathbf{u}|^2 - \mathbf{u} \cdot (\mathbf{v}_1 + \mathbf{v}_2) + \tfrac{1}{4}|\mathbf{v}_1 + \mathbf{v}_2|^2 = \tfrac{1}{4}|\mathbf{v}_1 - \mathbf{v}_2|^2.$$

Since $\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$, both $|\mathbf{v}_1 + \mathbf{v}_2|^2$ and $|\mathbf{v}_1 - \mathbf{v}_2|^2$ equal $2k^2$, so these cancel and $|\mathbf{u}|^2 = \mathbf{u} \cdot (\mathbf{v}_1 + \mathbf{v}_2)$. With $\mathbf{u} = \alpha(\mathbf{v}_1 + \mathbf{v}_2)$ this is

$$2\alpha^2 k^2 = 2\alpha k^2 \quad \Longrightarrow \quad \alpha(\alpha - 1) = 0.$$

So $\alpha = 0$, giving $\mathbf{u} = \mathbf{0}$, or $\alpha = 1$, giving $\mathbf{u} = \mathbf{v}_1 + \mathbf{v}_2$. Hence if $\mathbf{u} \neq \mathbf{0}$ then $\mathbf{u} = \mathbf{v}_1 + \mathbf{v}_2$, as required.

This makes sense physically. If C is at rest, the relative velocities are $\mathbf{v}_1$ and $\mathbf{v}_2$ themselves, which already satisfy the conditions. If $\mathbf{u} = \mathbf{v}_1 + \mathbf{v}_2$, the relative velocities are $-\mathbf{v}_2$ and $-\mathbf{v}_1$, which again are equal in length and perpendicular.

Checks

Take $\mathbf{v}_1 = (3,1)$, $\mathbf{v}_2 = (1,3)$, as in the diagram. The formula gives $\mathbf{u} = (2,2) \pm (1,1)$, that is $(3,3)$ or $(1,1)$. For $\mathbf{u} = (3,3)$ the relative velocities are $(0,-2)$ and $(-2,0)$; for $(1,1)$ they are $(2,0)$ and $(0,2)$. Both pairs are equal in length and perpendicular. A computer testing 10^4 random pairs $\mathbf{v}_1, \mathbf{v}_2$ confirmed the two identities and the two solutions, and for 10^3 random equal perpendicular pairs it found the solutions to be exactly $\mathbf{0}$ and $\mathbf{v}_1 + \mathbf{v}_2$.

Common slips

- Writing the relative velocity of A to C as $\mathbf{u} - \mathbf{v}_1$. The sign doesn't affect these conditions, but it matters in most relative-velocity questions.
- Forgetting to say why there are *exactly* two velocities. The count depends on working in a plane (in three dimensions the solutions form a whole circle) and on the radius being non-zero.

Beyond STEP

Moving from $\mathbf{u}$ to $\mathbf{w} = \mathbf{u} - \frac{1}{2}(\mathbf{v}_1 + \mathbf{v}_2)$ is a change of reference frame, and the fact that relative velocities don't change when every velocity is shifted by the same vector is

Galilean invariance, the starting point of classical mechanics. The "velocity diagram" picture used here, with velocities drawn as points, also turns collision and pursuit problems into geometry. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops relative motion and reference frames carefully.

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